1. Basic Probability
2. Interpretations of Probability
3. Basic probability of counting
4. Conditional Probability
5. Total probability
6. Bayes theorem

Probability

1. Study of randomness and uncertainty
2. Methods for quantifying chances

Random experiment

An experiment that can result in different outcomes, even though it is repeated in the same manner every time.

Example:

1. Tossing a coin
2. Throwing a die

Sample space

The set of all possible outcomes of an experiment. The sample space is denoted by S.

Examples

1. Tossing a coin, $S = \{H, T\}$.
2. Throwing a die, $S = \{1, 2, 3, 4, 5, 6\}$.
3. Throwing a die and tossing a coin simultaneously, $S = \{(1, H), (2, H), (3, H), (4, H), (5, H), (6, H), (1, T), (2, T), (3, T), (4, T), (5, T), (6, T)\}$

Observe: The usage of the Cartesian product.

Event

An event is a subset of the sample space of a random experiment.

Example: Let, $S = \{HH, HT, TH, TT\}$
$E_1 = \{HH, HT, TH\}$, $E_2 = \{TT\}$

Mutually Exclusive events

Two events E_1 and E_2 are mutually exclusive if $E_1 \cap E_2 = \phi$.

Example: Let $S = \{HH, HT, TH, TT\}$
$E_1 = \{HH\}$, $E_2 = \{TT\}$ then $E_1 \cap E_2 = \phi$.

Problem 1:

There are 5 red balls, 3 blue balls, and 1 green ball in an urn. Two balls are drawn randomly, one after another, without replacing the first ball. Write down the sample space.

Problem 2:

There are 5 red balls, 3 blue balls, and 1 green ball in an urn. Two balls are drawn simultaneously. Write down the sample space.

Problem 3:

Consider an urn containing 4 red balls, 2 blue balls, and 5 green balls. Three balls are drawn at random from the urn. Write down the event that the number of green balls exceeds the number of red balls.

Problem 1 - Ans:

$S = \{RR, RB, RG, BB, BR, BG, GR, GB\}$

Problem 2 - Ans:

$S = \{RR, RB, RG, BB, BG\}$

Problem 3 - Ans:

$S = \{RRR, RRB, RRG, RBB, RGG, BBG, BGG, GGG, RBG\}$

Desired event = {RGG, BBG, BGG, GGG, GGB}

Probability (Mathematical definition)

Let S be the sample space and A be an event associated with a random experiment. Let $n(S)$ and $n(A)$ be the number of elements of S and A. Then probabaility of event A occurring, denoted as $P(A)$, is defined by

$$P(A) = \frac{n(A)}{n(S)} = \frac{\textit{Number of cases favourable to A}}{\textit{Exhaustive number of cases in S}}$$

Examples

What is the probability of getting an even number in the die tossing experiment?
$S = \{1, 2, 3, 4, 5, 6\}$, $E = \{2, 4, 6\}$, $n(S) = 6$, $n(E) = 3$,
$P(A) = \frac{3}{6} = \frac{1}{2}$.

Statistical definition of probability

Let a random experiment be repeated n times and let an event A occur n_A times out of the n trials. The ratio $\frac{n_A}{n}$ is called the relative frequency of the event A. As n increases, $\frac{n_A}{n}$ shows a tendency to stabilise and to approach a constant value. This value, denoted by $P(A)$ is called the probability of the event A, i.e., $P(A) = \lim_{n\to\infty} \frac{n_A}{n}$.

Example

In a factory, a machine produces a large number of items. Over a long period, records show that out of 10000 items produced, 500 are found to be defective. If one item is chosen at random from the production, what is the probability that it is defective?

$$P(defective\ item) = \frac{Number\ of\ defective\ item}{Total\ number\ of\ items}$$

Axiomatic definition of Probability

Let S be the sample space and A be an event associated with a random experiment. Then the probability of the event A, denoted by $P(A)$, is defined as a real number satisfying the following axioms.

i) $0 \leq P(A) \leq 1$

ii) $\mathrm{P(S)} = 1$

iii) If A and B are mutually exclusive events then $P(A \cup B) = P(A) + P(B)$

iv) If $A_1, A_2, A_3, ...A_n, ...$ are a set of mutually exclusive events then $P(A_1 \cup A_2 \cup ... \cup A_n) = P(A_1) + P(A_2) + ... + P(A_n) + ...$

Theorem 1

The probability of the impossible event is zero.

Theorem 2

If $\bar{A}$ is the complementary event of A, $P(\bar{A}) = 1 - P(A) \leq 1$.

Theorem 3

If A and B are any 2 events, then
$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$.

Theorem 4

If $B \subset A$, $\quad P(B) \leq P(A)$.

1 Basic Probability

2 Interpretations of Probability

3 Basic probability of counting

4 Conditional Probability

5 Total probability

6 Bayes theorem

Interpretations of Probability

Probability can be interpreted in different ways depending on the context and the nature of uncertainty involved. The two commonly used interpretations are:

a) Subjective Interpretation views probability as a measure of personal belief or confidence regarding the occurrence of an event.

- It is a **personal interpretation** of probability.
- Probability represents a person's **degree of belief** in the occurrence of an event.
- The value of probability may vary from person to person based on:
  - prior knowledge,
  - experience,
  - available information.
- It does not require repeated experiments.

Frequency Interpretation of Probability

b) Frequency Interpretation defines probability in terms of long-run relative frequency.

- Probability is viewed as a **relative frequency**.
- It is based on the **proportion of times** an event occurs in repeated identical trials.
- Probability is considered an **objective property of that outcome**.
- This interpretation requires:
  - a large number of repetitions,
  - identical experimental conditions.

Examples

1. **Examples of Subjective Probability**
   - Probability that it will rain tomorrow.
   - Probability that a particular team will win a match.
   - Probability that a candidate will pass an examination.

In such situations, probabilities are assigned based on personal judgment rather than long-run experimental data.

2. **Examples of Frequency Probability**
   - Probability of getting a head when a fair coin is tossed.
   - Probability of getting a particular number when a die is thrown.
   - Probability of drawing a red card from a well-shuffled deck.

These probabilities can be verified experimentally by repeating the experiment a large number of times.

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Problem 1

Two balls are randomly drawn from a bowl containing 6 white and 5 black balls. What is the probability that one of the drawn balls is white and the other black?

Problem 2

Mr. Jones has 10 books that he is going to put on his bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Jones wants to arrange his books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

Problem 3

A class in probability theory consists of 6 men and 4 women. An exam is given and the students are ranked according to their performance. Assuming that no two students obtain the same score, (a) how many different rankings are possible? (b) If all rankings are considered equally likely, what is the probability that women receive the top 4 scores?

Problem 4:

A fair coin is tossed 4 times. Write the Sample Space clearly, give the table of outcomes corresponding to the following events and find the respective probabilities.
(A) More heads than tails are obtained
(B) Tails occur on the even numbered tosses

Problem 5:

There are 4 letters and 4 envelopes. If the letters are placed in the envelopes at random, find the probability that

i) None of the letters is in the correct envelope

ii) At least one letter is in the correct envelope.

Problem 6:

A lot consists of 10 good articles, 4 with minor defects and 2 with major defects. Two articles are taken from the lot at random (without replacement). Find the probability that
(i) both are good
(ii) both have major defects
(iii) at least 1 is good
(iv) at most 1 is good
(v) exactly 1 is good
(vi) neither has major defects
(vii) neither is good.

Problem 7:

From 6 positive and 8 negative numbers, 4 numbers are chosen at random (without replacement) and multiplied. What is the probability that the product is positive?

Problem 8:

A box contains tags marked $1, 2, \ldots, n$. Two tags are chosen at random without replacement. Find the probability that the numbers on the tags will be consecutive integers.

Problem 9:

If n biscuits are distributed at random among m children, what is the probability that a particular child receives r biscuits (where $r < n$)?

Problem 10:

A bag contains 5 red balls, 6 blue balls and 8 black balls. A ball is drawn at random. Find the probability that it is
(i) Red
(ii) Not blue
(iii) Blue or black

Problem 11:

An urn contains 3 red balls, 5 black balls and 7 blue balls. Two balls are selected at random. Find the probability that

1. both are red
2. one is black and the other is blue
3. neither of them is red
4. at least one is red

Problem 12:

An urn contains 3 red balls, 5 black balls and 7 blue balls. Nine balls are drawn at random. Find the probability that

1. 2 red balls, 3 black balls, and 4 blue balls are drawn
2. none of them is red
3. exactly one of them is red
4. at least one of them is red

Problem 13:

Three dice are rolled. Find the probability that the same number appears on exactly two of the three dice.

Problem 14:

From a selection of 38 battery cells containing 5 defective ones, we select 4 cells at random for quality check. Find the probability that half of them would be defective.

Problem 15:

A bag contains tickets numbered from 1 to 50. What is the probability that a randomly drawn ticket is a multiple of either 6 or 4?

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Conditional Probability

The conditional probability of an event B, assuming that the event A has happened, is denoted by $P(B/A)$ and defined as

$$P(B/A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) \neq 0$$

Example

A fair dice is tossed, Given that an odd number has been obtained, find the conditional probability of getting 1.

$$S = \{1, 2, 3, 4, 5, 6\}; \quad A = \{1, 3, 5\}; \quad B = \{1\}$$

$$\therefore \quad P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{3}$$

Product Theorem of Probability

Rewriting the definition of conditional probability, we get

$$P(A \cap B) = P(A) \times P(B/A).$$

Proof:

Let n_A, n_{AB} be the number of cases favourable to the events A and $A \cap B$, out of the total number n of cases.

$$\therefore \quad P(A \cap B) = \frac{n_{AB}}{n} = \frac{n_A}{n} \times \frac{n_{AB}}{n_A} = P(A) \times P(B/A).$$

The product theorem can be extended to 3 events A, B and C as follows:

$$P(A \cap B \cap C) = P(A) \times P(B/A) \times P(C/A \text{ and } B)$$

Properties

The following properties are easily deduced from the definition of conditional probability:

1. If $A \subset B$, $P(B/A) = 1$, since $A \cap B = A$
2. If $B \subset A$, $P(B/A) \geq P(B)$, since $A \cap B = B$, and

$$\frac{P(B)}{P(A)} \geq P(B),$$

   as $P(A) \leq P(S) = 1$
3. If A and B are mutually exclusive events, $P(B/A) = 0$, since $P(A \cap B) = 0$
4. If $P(A) > P(B)$, $P(A/B) > P(B/A)$
5. If $A_1 \subset A_2$, $P(A_1/B) \leq P(A_2/B)$

Problem: 1

A card is drawn at random from a well-shuffled pack of 52 playing cards. Find the probability that the card is an ace, given that it is a spade.

Problem: 2

In a class, 60% of the students pass Mathematics, 50% pass Physics, and 40% pass both subjects. Find the probability that a student passes Mathematics, given that the student has passed Physics.

Problem: 3

A box contains 4 bad and 6 good tubes. Two are drawn out from the box at a time. One of them is tested and found to be good. What is the probability that the other one is also good?

Independent event

Definition

A set of events is said to be independent if the occurrence of any one of them does not depend on the occurrence or non-occurrence of the others.

Why do we get the product $P(A)P(B)$?

Let, Two events A and B are independent, it is obvious from the definition that $P(A/B) = P(A)$. Since, $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

Then $P(A) = \frac{P(A \cap B)}{P(B)} \implies P(A \cap B) = P(A)\,P(B)$.

The product theorem can be extended to any number of independent events: If $A_1, A_2, A_3, \ldots A_n$ are n independent events.
$P(A_1 \cap A_2 \cap A_3 \cap \ldots \cap A_n) = P(A_1) \times P(A_2) \times P(A_3) \times \ldots \times P(A_n)$.

Theorem 1

If the events A and B are independent, then A and $\bar{B}$ are also independent.

Theorem 2

If the events A and B are independent, then $\bar{A}$ and B are also independent.

Theorem 3

If the events A and B are independent, then $\bar{A}$ and $\bar{B}$ are also independent.

Observe

$P(A \cup B) = 1 - P(\bar{A}) \times P(\bar{B})$

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Theorem of Total Probability

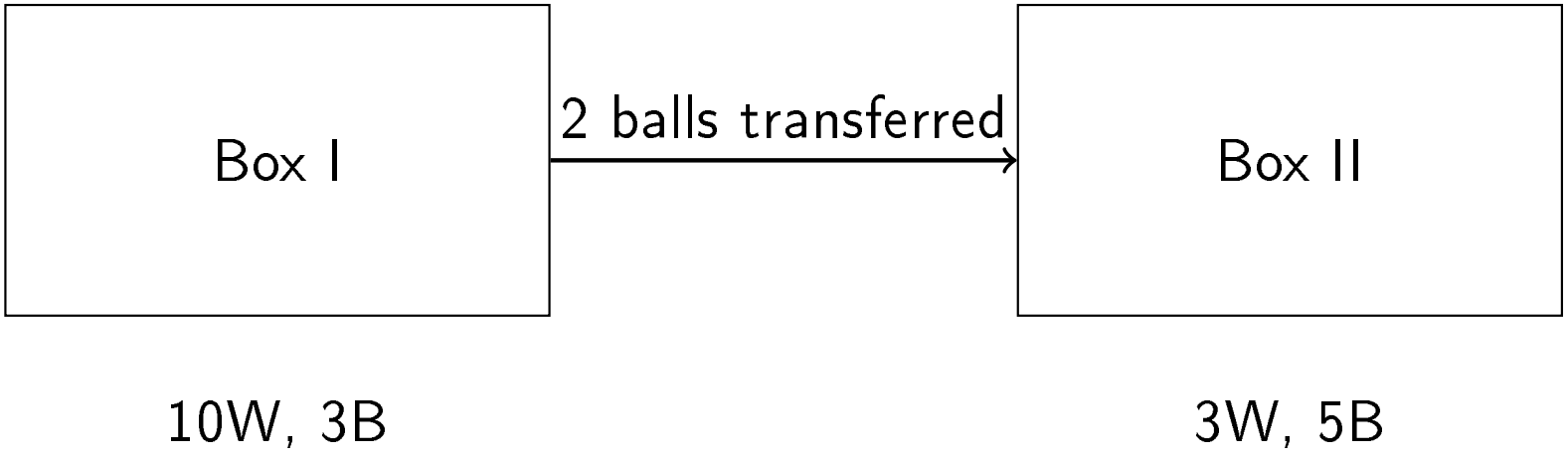
If $B_1, B_2, \ldots, B_n$ are mutually exclusive and exhaustive events and A is another event associated with B_i then

$$P(A) = \sum_{i=1}^{n} P(B_i)\, P(A \mid B_i)$$

Used when: An event can occur via multiple cases.

Problem 1

A box contains 10 white balls and 3 black balls. Another box contains 3 white balls and 5 black balls. Two balls are drawn at random from the first box and placed into the second box. Then one ball is drawn at random from the second box. What is the probability that this ball is white?



Let:

- A_1: Two white balls transferred
- A_2: Two black balls transferred
- A_3: One white and one black transferred

These events are mutually exclusive and exhaustive.

$$P(A_1) = \frac{{}^{10}C_2}{{}^{13}C_2} = \frac{15}{26}$$

$$P(A_2) = \frac{{}^{3}C_2}{{}^{13}C_2} = \frac{1}{26}$$

$$P(A_3) = \frac{{}^{10}C_1 \cdot {}^{3}C_1}{{}^{13}C_2} = \frac{5}{13}$$

Case	Box II after transfer
A_1	5W, 5B
A_2	3W, 7B
A_3	4W, 6B

Let B = drawing a white ball from Box II.

$$P(B \mid A_1) = \frac{5}{10}; \quad P(B \mid A_2) = \frac{3}{10}; \quad P(B \mid A_3) = \frac{4}{10}$$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$= \frac{15}{26} \cdot \frac{5}{10} + \frac{1}{26} \cdot \frac{3}{10} + \frac{5}{13} \cdot \frac{4}{10}$$

$$\boxed{P(\text{white ball}) = \frac{59}{130}}$$

Problem 2

A blot is manufactured by 3 machines A, B and C. A turns out twice as many items as B and machines B and C produce equal number of items. 2% of blots produced by A and B are defective and 4% of bolts produced by C are defective. All blots are put into 1 stock pile and 1 is chosen from this pile. What is the probability that it is defective?

Problem 3

In a coin tossing experiment, if the coin shows head, 1 dice is thrown and the result is recorded. But if the coin shows tail, 2 dice are thrown and their sum is recorded. What is the probability that the recorded number will be 2?

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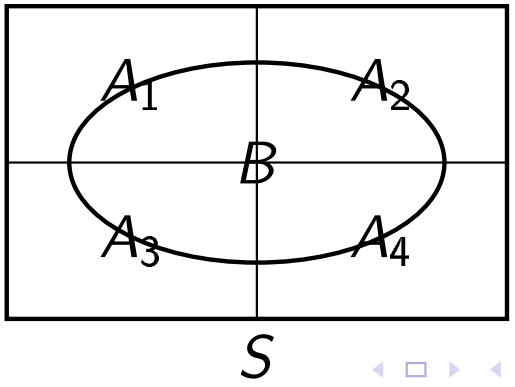
Bayes Theorem

Let $A_1, A_2, A_3, \ldots, A_n$ be a set of mutually exclusive and exhaustive events that together form the sample space S. Let B be any event from the same sample space such that $P(B) > 0$. Then,

$$P(A_k \mid B) = \frac{P(A_k \cap B)}{P(A_1 \cap B) + P(A_2 \cap B) + \cdots + P(A_n \cap B)}$$

$$= \frac{P(B \mid A_k)P(A_k)}{P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \cdots + P(B \mid A_n)P(A_n)}$$

Let $n = 4$



Problem 1

Three companies X, Y, and Z supply computers to a university.

Company	Percentage Supplied	Probability of Defective
X	60%	0.01
Y	30%	0.02
Z	10%	0.03

Given that a computer is defective. Find the probability that it was supplied by Y?

Answer:

$$P(Y \mid D) = \frac{P(Y \cap D)}{P(X \cap D) + P(Y \cap D) + P(Z \cap D)}$$

First, compute the joint probabilities:

$$P(Y \cap D) = P(D \mid Y)\, P(Y) = 0.02 \times 0.3 = 0.006$$

$$P(X \cap D) = P(D \mid X)\, P(X) = 0.01 \times 0.6 = 0.006$$

$$P(Z \cap D) = P(D \mid Z)\, P(Z) = 0.03 \times 0.1 = 0.003$$

Now substitute:

$$P(Y \mid D) = \frac{0.006}{0.006 + 0.006 + 0.003} = \frac{0.006}{0.015} = \frac{6}{15} = \frac{2}{5}$$

$$\boxed{P(Y \mid D) = \frac{2}{5}}$$

Problem 2

A bag contains 5 balls and it is not known how many of them are white. Two balls are drawn at random from the bag and they are noted to be white. What is the chance that all the balls in the bag are white?

Problem 3

There are 3 true coins and 1 false coin with 'head' on both sides. A coin is chosen at random and tossed 4 times. If 'head' occurs all the 4 times, what is the probability that the false coin has been chosen and used?

Problem 4

In a college, students are classified into three groups based on their attendance, namely high attendance, medium attendance, and low attendance. It is known that 50% of the students have high attendance, 30 % have medium attendance, and the remaining 20% have low attendance. Past records show that 5% of the students with high attendance fail the examination, 10% of those with medium attendance fail, and 25% of those with low attendance fail.

A student is selected at random and is found to have failed the examination. What is the probability that the student had low attendance?

Thanks!